

ABEE AGDM, AB, Canada

WMO: 712850

Lat: 54.2773N

Lon: 112.9654W

Elev: 2178

StdP: 13.58

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.1	-22.2	-35.7	0.8	-29.1	-28.3	1.2	-21.7	21.0	26.8	18.6	26.4	3.2	140	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	81.1	63.2	78.0	61.7	75.0	60.3	66.2	77.2	64.1	74.5	62.1	71.8	8.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	89.9	71.9	59.9	83.4	69.3	57.9	77.6	66.8	32.1	77.1	30.4	74.8	28.9	71.7	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.8	14.8	DB	-39.3	86.2	6.4	2.5	-44.0	88.0	-47.7	89.5	-51.4	90.9	-56.0	92.8	(4)
(5)			WB	-39.5	70.7	6.4	2.8	-44.1	72.7	-47.9	74.4	-51.5	76.0	-56.1	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	8.7	9.3	21.0	36.7	49.0	56.3	60.8	57.8	48.8	36.7	20.1	8.3
	DBStd	22.07	15.32	14.30	13.40	9.54	7.99	5.31	4.87	5.76	7.73	8.52	12.58	14.19
	HDD50	6576	1280	1139	899	405	116	10	0	7	112	417	898	1293
	HDD65	11129	1745	1559	1364	848	498	265	150	231	486	877	1348	1758
	CDD50	954	0	0	0	7	84	199	334	248	77	5	0	0
	CDD65	30	0	0	0	0	1	3	18	7	1	0	0	0
	CDH74	851	0	0	0	1	66	118	361	222	80	3	0	0
	CDH80	140	0	0	0	0	8	12	72	34	14	0	0	0
	WSAvg	6.7	6.3	6.3	7.3	8.2	7.9	6.9	6.2	5.9	6.4	7.0	6.5	5.8
	PrecAvg	15.9	0.7	0.5	0.6	0.9	1.6	2.8	3.1	2.0	1.3	0.7	0.6	0.7
(15) Precipitation	PrecMax	20.0	1.5	1.1	1.3	2.2	3.4	5.4	6.1	3.5	2.9	2.2	1.6	1.5
	PrecMin	11.8	0.0	0.0	0.1	0.1	0.2	0.9	0.9	0.8	0.4	0.2	0.2	0.0
	PrecStd	2.4	0.4	0.3	0.4	0.5	0.7	1.2	1.1	0.7	0.7	0.4	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	43.8	49.8	71.2	80.5	81.5	86.1	84.2	81.8	71.3	52.1	42.6
		MCWB	36.9	36.5	40.4	50.9	56.9	61.4	68.3	63.7	60.8	53.3	43.5	36.6
	2%	DB	39.0	38.2	44.9	64.1	75.1	77.3	81.6	79.7	75.4	60.8	43.7	37.4
		MCWB	34.0	33.1	37.5	47.6	54.4	60.0	65.8	63.3	58.8	48.4	37.5	32.8
	5%	DB	35.3	34.3	41.1	58.2	70.5	73.9	77.9	75.9	70.0	55.4	39.8	33.1
		MCWB	31.7	30.7	35.3	44.2	52.2	58.3	63.8	61.8	56.0	44.7	35.2	29.9
	10%	DB	31.0	30.5	38.1	53.1	66.8	70.2	74.7	72.7	64.6	51.0	35.6	28.7
		MCWB	28.4	27.8	33.4	41.8	50.3	56.4	62.2	60.7	53.2	42.5	32.4	26.2
	0.4%	WB	37.4	36.9	41.4	51.8	58.4	64.8	70.6	67.9	62.9	54.3	43.3	36.7
		MCDB	42.9	43.0	47.6	69.2	75.1	77.3	81.8	77.8	77.2	69.7	51.0	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.5	33.5	37.9	47.9	55.7	62.1	67.5	65.2	59.7	49.6	38.4	33.0
		MCDB	38.5	37.4	43.8	62.6	70.7	72.6	78.7	76.0	72.8	59.6	42.8	36.8
	5%	WB	31.7	30.8	35.7	44.9	53.6	60.0	65.3	63.1	57.0	45.9	35.5	30.1
		MCDB	35.4	34.2	40.6	56.5	67.5	70.7	75.7	73.4	69.0	54.2	39.0	33.1
	10%	WB	28.5	27.7	33.5	42.3	51.6	58.1	63.2	61.1	54.4	43.4	32.4	26.5
		MCDB	30.9	30.6	37.6	52.2	64.2	67.9	72.5	71.3	63.1	49.7	35.2	28.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	21.5	19.5	21.3	25.1	22.6	23.2	24.4	24.4	20.9	17.0	18.9
		MCDBR	21.9	25.8	21.9	31.6	33.1	30.2	30.1	31.3	35.5	29.7	21.4	22.3
		MCWBR	17.6	21.3	15.7	17.1	15.5	15.0	16.1	16.7	19.5	18.3	15.5	18.7
	5% WB	MCDBR	21.3	24.8	20.6	29.4	29.6	26.0	27.3	27.8	32.2	28.4	19.7	22.2
		MCWBR	17.5	20.6	15.0	16.9	14.3	13.8	15.5	15.8	18.1	17.9	14.9	18.8
(40) Clear-Sky Solar Irradiance	taub		0.253	0.271	0.305	0.325	0.338	0.352	0.351	0.359	0.305	0.280	0.249	0.235
	taud		2.280	2.295	2.304	2.386	2.406	2.378	2.430	2.354	2.516	2.557	2.455	2.289
	Ebn at Noon		228	264	278	287	288	284	282	272	278	263	236	212
	Edh at Noon		18	25	31	33	34	36	33	34	25	19	15	14
(44) All-Sky Solar Radiation	RadAvg		274	599	1079	1477	1802	1886	1909	1565	1080	607	284	199
	RadStd		31	43	81	123	147	126	101	111	111	80	31	17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page